

Roll No.:

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech(Biotech)

Semester:

Name of the Course: Statistical Data Analysis with R

Course Code: TCS-471

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)																			
(a)	A restaurant owner ranked his 17 waiters in terms of their speed and efficiency on the job. He correlated these ranks with the total amount of tips each of these waiters received for a one-week period. The obtained value of correlation coefficient is 0.438. What do you conclude?	CO1																		
(b)	The odds against Manager X settling the wage dispute with the workers are 8:6 and the odds in favor of manager Y settling the same dispute are 14:16. (i) What is the chance that neither settles the dispute, if they both try, independently of each other? (ii) What is the probability that the dispute will be settled?	CO1 CO1																		
(c)	The odds that a book will be favourably reviewed by 3 independent critics are 5 to 2, 4 to 3 and 3 to 4 respectively. What is the probability that, of the three reviews, a majority will be favourable?																			
Q2																				
(a)	The mean height of 50 male students who showed above average participation in college athletics was 68.2 inches with a standard deviation of 2.5 inches; while 50 male students who showed no interest in such participation had a mean height of 67.5 inches with a standard deviation of 2.8 inches. (i) Test the hypothesis that male students who participate in college athletics are taller than other male students. (ii) By how much should the sample size of each of the two groups be increased in order that the observed difference of 0.7 inches in the mean heights be significant at the 5% level of significance.	CO2																		
(b)	A Poisson distribution has a double mode at $x = 1$ and $x = 2$. What is the probability that x will have one or the other of these two values?	CO2																		
(c)	Compute the best fit exponential curve of the form $Y = ab^x$ to the following dataset <table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>Y</td><td>1.0</td><td>1.2</td><td>1.8</td><td>2.5</td><td>3.6</td><td>4.7</td><td>6.6</td><td>9.1</td></tr></table>	X	1	2	3	4	5	6	7	8	Y	1.0	1.2	1.8	2.5	3.6	4.7	6.6	9.1	CO2
X	1	2	3	4	5	6	7	8												
Y	1.0	1.2	1.8	2.5	3.6	4.7	6.6	9.1												
Q3																				
(a)	List and define some basic data structures in R.	CO3																		
(b)	How to import data in R?	CO3																		

(c)	(i) How to create a data frame in R? (ii) How do you add a new column to a data frame in R?	CO4
Q4		
(a)	What is the difference between the functions <code>apply()</code> , <code>lapply()</code> , <code>sapply()</code> , and <code>tapply()</code> ?	CO4
(b)	What are correlation and covariance, and how do you calculate them in R?	CO5
(c)	(i) What is Shiny in R? (ii) What is the difference between the <code>with()</code> and <code>within()</code> functions?	CO5
Q5		
(a)	List down at least five libraries in R that can be used for data visualization. Explain three of them briefly.	CO6
(b)	How to make multiple plots on to a single page layout in R? Explain with an example.	
(c)	Provide three differences between <code>ggplot2</code> and <code>lattice</code> packages?	CO6
		CO6